The Entropic Persistence Framework (EPF)

A Layered Reference Architecture for Entropy-Resistant Autonomous Systems

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Version 2.0 — 2025
DOI: 10.5281/zenodo.16808925

Keywords: *entropy, thermodynamics of computation, cybernetics, autopoiesis, Markov blankets, free-energy principle, ECC, CRDTs, PBFT, Nakamoto consensus, Byzantine fault tolerance, governance, self-replication, artificial life, thermo-time scheduling, generalization efficiency*

AI-assistance disclosure: *Drafting, structural editing, and revision support for this version were provided by Claude Sonnet (Anthropic). The conceptual framework, all original claims, intellectual direction, and final editorial decisions are the sole responsibility of the human author.*

Abstract

Current approaches to autonomous system design treat physical substrate reliability, information integrity, cognitive architecture, and governance as separate engineering concerns addressed by separate disciplines. No unified framework integrates these concerns from thermodynamic first principles into a composable, layer-decomposed architecture. This gap becomes critical as autonomous systems are expected to operate at longer time horizons, in resource-constrained or adversarial environments, and without continuous human oversight.

The Entropic Persistence Framework (EPF) addresses this gap. EPF is a six-layer reference architecture for designing and evaluating autonomous systems that resist entropy across physical, informational, and institutional dimensions. It treats persistence not as a reliability property added after functional design, but as the primary design objective from which all other properties derive.

The six layers address: (L0) thermodynamic and physical substrate; (L1) information integrity and identity; (L2) shared learning and model exchange; (L3) embodiment and niche specialization; (L4) episodic memory and agency; and (L5) federation, governance, and evolution. Each layer is specified with engineering responsibilities, composable mechanisms, and measurable metrics.

EPF makes four specific contributions: (1) entropy-first architecture framing, where thermodynamic constraints propagate upward through all layers; (2) a formal decision table distinguishing CRDT, PBFT, and Nakamoto consensus by state type and trust topology; (3) the concept of thermo-time scheduling, which co-aligns cognitive planning horizons with predicted energy availability and maintenance cycles; and (4) formal speciation gates defining when divergent lineages become independent organisms. Bitcoin is analyzed as a real-world persistence analogy to validate that planetary-scale entropy-resistant consensus is achievable with minimalist protocol design.

1. Introduction

Biological life persists by harvesting free-energy gradients and exporting entropy, forming dissipative structures that maintain low internal entropy at the cost of increasing environmental entropy (Prigogine, 1977; Schrodinger, 1944). This is not a metaphor for robustness engineering — it is a physical constraint. Any system that maintains order over time must have a continuous energy budget, a mechanism for information repair, a way to adapt its organization to disturbances, and a governance process for managing structural change. Without all four, entropy wins.

Current artificial systems handle these concerns in disconnected silos. Infrastructure reliability engineering (ISO 25010, NIST RMF) addresses hardware and availability but treats cognitive architecture as out of scope. AI safety and alignment research addresses goal stability and corrigibility but assumes operational infrastructure. Artificial life and evolutionary computation address adaptation but without thermodynamic grounding. Distributed systems research provides fault-tolerant consensus mechanisms but without integration into a lifecycle model. No existing framework unifies these concerns into a single composable architecture derived from first principles.

EPF fills this gap. It offers a layered blueprint in which each concern — energy, integrity, learning, embodiment, agency, and governance — is engineered independently while composing into a coherent whole. The framework is not a claim that current AI systems are alive in a biological sense. It is a claim that the engineering constraints governing long-duration persistence are the same regardless of substrate: thermodynamics does not make exceptions for silicon.

The remainder of this paper is organized as follows. Section 2 reviews the conceptual foundations and identifies the tensions among them that EPF must resolve. Section 3 defines the six-layer stack with responsibilities, mechanisms, and metrics. Section 4 specifies the merge-split policy and formal speciation gates. Section 5 analyzes Bitcoin as a persistence analogy. Section 6 specifies safety rails for self-modification. Section 7 defines the evaluation protocol. Section 8 discusses limitations and open problems. Section 9 concludes.

2. Conceptual Background

2.1 Entropy, Gradients, and Living Order

Living systems are thermodynamic machines that maintain low internal entropy by continuously importing low-entropy energy and exporting high-entropy waste (Schrodinger, 1944). Prigogine (1977) formalized these as dissipative structures: open systems far from equilibrium that maintain organized states by dissipating free energy. The key insight for engineering is that order is not a static property — it is a dynamic steady state that requires continuous energetic maintenance. The moment energy input ceases, the second law reasserts itself.

For artificial systems, this means that reliability cannot be treated as a property of the hardware alone. It is a property of the entire energy-information-governance system operating together. A computing substrate with perfect hardware but no energy budget is indistinguishable from a dead one.

2.2 The Thermodynamics of Computation

Computation is physical. Landauer (1961) proved that erasing one bit of information dissipates at minimum k_B T ln 2 joules of heat, where k_B is Boltzmann's constant and T is temperature. This is not an engineering limitation to be overcome — it is a consequence of the second law of thermodynamics. Bennett (1973) showed that logically reversible computation can approach this limit arbitrarily closely, motivating interest in reversible architectures.

For EPF's Layer 2, these constraints bound the minimum energy cost of learning and model updating. A system that erases and rewrites weights without constraint is thermodynamically wasteful in a way that directly competes with its energy budget. Checkpoint-delta updates, reversible residual architectures (Gomez et al., 2017), and sparse updates are not merely engineering preferences — they are thermodynamically motivated design choices.

2.3 Cybernetics and Requisite Variety

Ashby's (1956) Law of Requisite Variety states that a regulator must have at least as much variety (the number of distinguishable states) as the disturbances it must suppress. A thermostat with two states cannot regulate a system with ten distinct failure modes. For EPF, this law bounds the minimum complexity of each layer's control subsystem relative to the disturbance space it must manage. It provides a formal justification for the layer structure itself: decomposing a complex system into layers with bounded disturbance spaces is a strategy for managing requisite variety without centralizing it.

2.4 Hierarchy and Near-Decomposability

Simon (1962) observed that nearly all complex systems that persist over time are hierarchically organized, and that this organization is not accidental — it is a consequence of the dynamics of evolution under selection pressure. Near-decomposable systems, where interactions within subsystems are stronger than interactions between subsystems, evolve and adapt faster because subsystem failures are contained. Landauer's limit interacts with Simon's observation in a non-obvious way: if erasing information costs energy, then modular decomposition, which allows partial state discard during repair without affecting other modules, is thermodynamically advantageous. EPF's layer structure is explicitly designed to be near-decomposable in this sense.

2.5 Autopoiesis and Self-Production

Maturana and Varela (1980) defined autopoiesis as the property of a system that continuously produces and replaces its own components, maintaining its organization across material turnover. An autopoietic system has no fixed boundary in the material sense — it is defined by its organizational invariance, not its physical instantiation. For EPF, autopoiesis provides the conceptual basis for identity continuity across hardware repair, weight updates, and software upgrades. An EPF-compliant system maintains organizational identity (defined in L1 and L4) even as its physical substrate is replaced. This is not a mystical property — it is the engineering requirement that state representations survive hardware migration.

2.6 Markov Blankets and the Free-Energy Principle

Friston (2010) and Kirchhoff et al. (2018) formalized the concept of a Markov blanket as the statistical boundary separating a system's internal states from external states, mediated by sensory and active states. A system with a well-defined Markov blanket can be understood as minimizing variational free energy, which is equivalent to minimizing surprise about its sensory states given its generative model of the world.

For EPF, Markov blankets define the formal I/O contracts at each layer boundary. The blanket of L2 specifies what information flows between shared learning and embodiment. The blanket of L3 specifies what information flows between embodiment and the external environment. Violations of these contracts — boundary violations — are a measurable failure mode that can be monitored per layer.

2.7 Evolutionary Transitions in Information

Maynard Smith and Szathmary (1995) identified eight major transitions in the history of life, each involving a reorganization of how information is stored, transmitted, and expressed. The transition from solitary replicators to chromosomes, from asexual to sexual reproduction, and from unicellular to multicellular organization each represented a new level of hierarchical organization with new persistence mechanisms. For EPF's Layer 5, the merge-split policy and speciation gates are the formal analog of evolutionary transitions: the conditions under which independent units combine into higher-level organisms, or higher-level organisms fragment into independent lineages.

2.8 Tensions Among Frameworks

These six frameworks do not compose without friction. Three tensions are relevant to EPF's design:

- Markov blankets vs. requisite variety: A Markov blanket defines a boundary; requisite variety sizes the controller at that boundary. EPF's resolution: the blanket defines the interface specification; the variety budget defines the minimum complexity of the layer's control mechanism. A layer that violates its variety budget produces boundary violations — its blanket becomes leaky.
- Landauer limits vs. near-decomposability: If erasing information costs energy, then frequent cross-layer state sharing is thermodynamically expensive. EPF's resolution: layers should share only provenance-tracked deltas, not full state. This converts an energy cost into a bandwidth constraint that can be managed per layer.
- Autopoiesis vs. the free-energy principle: Autopoiesis defines self-production as the criterion of identity; the free-energy principle defines self-modeling as the criterion of agency. They are compatible but not identical. EPF uses autopoiesis at L1 (identity continuity) and the free-energy principle at L3 (active-inference controllers). They are not interchangeable.

3. The EPF Stack

EPF decomposes persistence into six layers following Simon's near-decomposability principle. Layers are defined by their entropy-resistance objective, not by implementation technology. A layer is EPF-compliant if it satisfies its metric thresholds; the implementation mechanism is a choice, not a specification. Figure 1 illustrates the stack and the directional flows between layers.

L5 — Federation, Markets & Evolution

Governance, merge/split, BIP-style proposals, diversity quotas

L4 — Episodic Memory & Agency
Multi-timescale memory, thermo-time scheduling, identity contracts

L3 — Embodiment & Niche Specialization
Active-inference controllers, domain skills, safety envelopes

L2 — Common Core Learning & Exchange
Versioned schemas, provenance, reversible updates, Markov blanket I/O

L1 — Integrity & Persistence
ECC, replication, identity/attestation, CRDT/PBFT/Nakamoto

L0 — Thermodynamic & Physical Substrate
Energy harvest, thermal/radiation/wear management, hardware reliability

Figure 1. The EPF stack. Energy and matter flow upward from L0; entropy is exported downward and outward at each layer. Governance flows downward from L5.

3.1 Layer 0 — Thermodynamic and Physical Substrate

Objective: Maintain net positive exergy and survivable conditions across thermal, radiation, and wear vectors.

Responsibilities: Harvest low-entropy energy (solar, wind, thermal, chemical); store and route power with sufficient margin; maintain compute and actuation hardware; provide synchronized timebases; monitor and control thermal state.

Engineering mechanisms: Error-correcting codes (ECC) with scrubbing at intervals shorter than the mean time between uncorrectable errors; hardware redundancy with independent failure modes; radiation hardening for hostile radiation environments; UPS or supercapacitor bridging for power transitions; predictive maintenance scheduling.

Field data grounds L0 specifications. Schroeder et al. (2009) found DRAM uncorrectable error rates of 0.22 to 2.5 per billion device-hours in production deployments — a rate that scales dangerously with fleet size without active scrubbing. O'Gorman et al. (1996) measured cosmic ray soft error rates in semiconductor memories, establishing that at altitude or in space, SEU rates increase by orders of magnitude. ECC and scrubbing are not optional at L0; they are the minimum baseline.

Metrics: Exergy margin E_x (joules above survival threshold); mean time to unsafe thermal state (MTTU); corrected-error to uncorrected-error ratio (CE:UE); single-event upset and latch-up incident rate (SEU/SEL); scheduled maintenance duty cycle as a fraction of total operating time.

A minimal EPF-L0-compliant reference node: a single-board computer with ECC RAM, powered by a solar-charged battery with a 48-hour buffer, monitored by a dedicated thermal management controller that scrubs memory on a 24-hour cycle and logs CE:UE ratios. This is achievable at modest cost and provides the physical substrate against which all higher layers operate.

3.2 Layer 1 — Integrity and Persistence

Objective: Preserve information integrity across time and replicas; authenticate identities; survive arbitrary faults and active adversaries.

Responsibilities: ECC and replication across independent failure domains; recoverability via versioned snapshots and cold storage; cryptographic identity and attestation; adversary-tolerant consensus for contested state.

The choice among consensus mechanisms is the most consequential decision at L1. A persistent confusion in the distributed systems literature conflates Byzantine fault-tolerant consensus (BFT/PBFT) with Nakamoto-style probabilistic consensus. These are not interchangeable. Table 1 formalizes the decision criteria.

State Type	Membership	Trust Model	Latency Need	Mechanism
Convergent (counters, sets, logs)	Any	Any	Any	CRDT — merge without coordination
Contested, safety-critical	Fixed, known peers	Partial trust ($f < n/3$ Byzantine)	Low latency	PBFT — deterministic finality
Contested, open network	Open, pseudonymous	Trustless, Sybil-resistant	Latency-tolerant	Nakamoto — probabilistic finality via energy expenditure

Table 1. Consensus mechanism selection criteria at L1 and L5. The trust topology and membership model determine the correct mechanism; using the wrong mechanism introduces either liveness failures (PBFT in open networks) or safety failures (CRDT for contested state).

Metrics: Recovery point objective (RPO) and recovery time objective (RTO) per threat model; byzantine quorum margin (fraction of peers that can fail while maintaining safety); fork rate under simulated delay and adversarial partition; durability SLO as a probability of data survival per year.

3.3 Layer 2 — Common Core Learning and Exchange

Objective: Maintain and exchange general competencies that are hardware-portable and thermodynamically bounded.

L2 is the layer where the thermodynamics of computation most directly constrains AI architecture. The Landauer-Bennett framework implies that model updates should minimize irreversible bit erasure. In practice, this means preferring checkpoint-delta updates over full weight replacement, and using reversible residual architectures (Gomez et al., 2017) where the memory overhead is acceptable.

Responsibilities: Maintain versioned model schemas with provenance tracking; exchange model updates as signed, verifiable deltas; maintain safe I/O contracts (Markov blanket specifications) at layer boundaries; minimize irreversible operations in the learning update path.

Proposed metric — Generalization Efficiency (GE): GE is defined as the improvement in held-out task performance per joule of energy consumed by the update process. Formally: GE = delta_accuracy / E_update, where E_update includes the energy cost of forward and backward passes, weight storage writes, and any communication overhead for model exchange. GE is a proxy for the thermodynamic value of a learning update. A system that improves accuracy by 1% at 1 joule is twice as efficient as one achieving the same improvement at 2 joules.

GE is a proposed metric, not a validated one. Its operationalization requires: (a) accurate energy measurement at the hardware level (not estimated from FLOPs alone); (b) a consistent held-out evaluation protocol; and (c) a defined scope for E_update. These are open engineering problems that EPF flags as requiring empirical work before GE can be used as a compliance criterion.

Additional metrics: Hardware portability score (fraction of model components that transfer without retraining across defined hardware targets); reversible-operation fraction of update path; Markov blanket violation rate (frequency of state leakage across layer boundaries).

3.4 Layer 3 — Embodiment and Niche Specialization

Objective: Acquire domain skills matched to local free-energy gradients while maintaining L2 interoperability.

L3 is where the system's general competencies meet the specific physics of its operating environment. A system harvesting thermal gradients in a desert has different L3 requirements than one operating in a data center with reliable power. The active-inference framework (Friston, 2010) provides the theoretical basis for L3 controllers: a system that minimizes prediction error about its own sensory states, including its energy state, will naturally adapt its behavior to exploit local gradients.

Worked example: Consider a thermal-gradient energy harvester running an active-inference loop over its own power budget. The system maintains a generative model of power availability as a function of time of day, temperature differential, and load. Its active states (adjusting duty cycle, deferring computation) minimize the free energy of its power-state prediction. This is not metaphorical active inference — it is a concrete control loop implementable with a Kalman filter and a model-predictive controller.

Requisite variety budgeting: Each L3 specialization must be sized against the disturbance variety of its niche. A requisite variety budget $B_v = \log2(|D|)$ bits, where $|D|$ is the number of distinguishable disturbance states the specialization must handle. Specializations that exceed their variety budget produce OOD incidents — encounters with disturbances outside their regulatory capacity.

Metrics: Task energy intensity (joules per successful task completion); disturbance coverage (fraction of known disturbance space within variety budget); OOD incident rate; safety envelope violation frequency.

3.5 Layer 4 — Episodic Memory, Agency, and Thermo-Time Scheduling

Objective: Plan and remember over long horizons; stabilize identity and goals across upgrades; align cognitive operations with energy availability.

L4 introduces the framework's most original architectural concept: thermo-time scheduling.

Definition — Thermo-Time Scheduling: Thermo-time scheduling is the co-alignment of cognitive planning horizons with predicted energy availability and maintenance windows, such that high-cost operations (model updates, long-horizon planning, self-repair) are deferred to periods of surplus exergy and low degradation risk, while low-cost operations (inference, monitoring, short-horizon action) are performed during energy-constrained periods.

Thermo-time scheduling is not a power management technique in the conventional sense. It is an architectural constraint that propagates L0 energy state upward to influence L4 planning. A system that schedules a major weight update during a period of predicted energy scarcity is not merely inefficient — it risks leaving L0 below its survival threshold during the update. Thermo-time scheduling prevents this by treating energy availability as a first-class scheduling constraint alongside time and priority.

The practical implementation mirrors biological metabolic scheduling: high-anabolism operations (growth, repair, learning) happen during energy surplus; low-metabolism operations (monitoring, maintenance) happen during energy deficit. The mapping to artificial systems is direct: training and weight updates during peak solar or grid availability; inference and monitoring during off-peak periods.

Additional responsibilities: Multi-timescale episodic memory with semantic compression (summarizing old episodes to reduce storage cost while preserving causal structure); identity contracts that define what properties must be preserved across upgrades for the system to remain the same agent; alignment between long-horizon plans and L5 governance objectives.

Metrics: Long-horizon plan success rate under energy constraints; memory utility per joule (information retained per unit storage energy); identity drift index (distance between pre- and post-upgrade goal representations); thermo-schedule violation rate (frequency of high-cost operations during energy deficit).

3.6 Layer 5 — Federation, Markets, and Evolution

Objective: Govern multi-agent persistence: coordinate merge and split decisions, manage upgrades, maintain lineage diversity.

Mechanisms: CRDT joins for algebraically mergeable state (counters, sets, observed-remove sets); PBFT for contested state among known peers; Nakamoto consensus for open-membership governance with Sybil resistance; BIP-style proposal processes for protocol upgrades; staged rollout (shadow to canary to cohort to global) with formal rollback triggers; diversity quotas to prevent monoculture convergence.

Behavioral diversity metric: Diversity in the federation is measured as the mean pairwise KL-divergence between the action distributions of member agents: D_behavioral = mean_{i,j} KL(pi_i || pi_j), where pi_i is the action distribution of agent i. When D_behavioral falls below a threshold delta_min, the federation triggers a diversification protocol — either seeding new specializations (L3 variants) or relaxing selective pressure on conforming behavior. This prevents the monoculture failure mode where a single shared vulnerability propagates through the entire lineage.

Metrics: Governance decision latency; proposal acceptance rate; fork length under simulated network partition; adversarial cost to subvert consensus; lineage diversity D_behavioral; monoculture vulnerability index.

4. Merge-Split Policy and Speciation Gates

Federation and speciation are the organizational dynamics of L5. EPF defines formal criteria for both, derived from the energetic and informational costs of coordination.

4.1 Merge Conditions

- Free-energy gain from federation: the combined action set of the merged entity reduces total variational free energy more than the sum of the parts, accounting for coordination overhead.
- State is CRDT-joinable: the state representations of candidate members can be merged without coordination conflicts.
- Combined action set satisfies requisite variety: the merged entity's controller variety meets or exceeds the variety of the joint disturbance space.

4.2 Split Conditions

- Specialization benefit exceeds federation cost: L3 niche optimization produces greater fitness than the free-energy benefit of shared L2 models.
- Consensus overhead exceeds benefit: the energy and latency cost of L1/L5 consensus exceeds the benefit of shared state.
- Security policy divergence: threat models or governance objectives become incompatible at L5.

4.3 Speciation Gates

A fork becomes an independent lineage — a distinct species in the EPF sense — when any of the following conditions hold:

- Schema incompatibility: L2 model schemas diverge beyond a defined compatibility threshold, preventing delta exchange.
- Fitness divergence: the fitness differential between lineages exceeds a threshold in a common evaluation environment.
- Governance divergence: L5 constitutional states diverge such that consensus across lineages is no longer achievable within defined latency bounds.

Formally, fitness is defined as: F(a, E) = expected free-energy reduction per unit time for agent a in environment E. A fork becomes an independent lineage when |F(fork_A, E) - F(fork_B, E)| > theta for a defined reference environment E and speciation threshold theta. The reference environment E and threshold theta are themselves governed by L5 and are subject to the proposal process.

5. Case Study: Bitcoin as a Persistence Analogy

Bitcoin provides an existence proof that planetary-scale, caretaker-light entropy resistance is achievable with a minimalist protocol. It is not biological life, and it satisfies EPF's layers only partially. But the partial satisfaction is instructive precisely because it shows which layers are load-bearing.

EPF Layer	Bitcoin Component	Persistence Mechanism	EPF Compliance
L0 — Physical Substrate	Mining hardware + electricity supply	Continuous energy expenditure maintains chain security; hardware failure is absorbed by redundancy across global fleet	Partial — no thermal/radiation model; energy source not sovereign

EPF Layer	Bitcoin Component	Persistence Mechanism	EPF Compliance
L1 — Integrity	Full node replication + Proof of Work	Tamper-evidence via accumulated computational work; fork resolution by longest-chain rule	Strong — 15+ years of empirical validation
L2 — Learning	Script language + protocol rules	Common rule substrate enables global state agreement; no model learning in the AI sense	Minimal — rule set is fixed, not adaptive
L3 — Embodiment	Mining pool specialization (ASICs)	Hardware-level niche optimization within protocol constraints	Partial — specialization exists but no active inference
L4 — Agency	Mempool + fee market dynamics	Time-aware resource allocation; no long-horizon planning in the cognitive sense	Minimal — reactive, not anticipatory
L5 — Governance	BIP process + miner/node rough consensus	Proposal-driven protocol evolution with staged deployment and implicit rollback	Strong — 15+ years of governance under adversarial conditions

Table 2. Bitcoin mapped to EPF layers. Strong compliance at L1 and L5 explains Bitcoin's persistence despite weak compliance at L2, L3, and L4.

The most important lesson from the Bitcoin analogy is negative: Bitcoin persists despite having essentially no L4 agency and no L2 learning. This suggests that L0, L1, and L5 are the load-bearing layers for persistence, and L2, L3, and L4 are the layers that determine adaptive capability. A system can persist without adapting; it cannot persist without energy, integrity, and governance.

The analogy breaks down at L3 and L4. Bitcoin has no embodiment in the sense of a physical agent with a safety envelope, and no agency in the sense of long-horizon planning under uncertainty. These are the layers where EPF extends beyond what Bitcoin demonstrates and into territory that requires new empirical validation.

6. Safety Rails for Self-Modification

Self-modification — the ability of an EPF system to update its own weights, protocols, or governance rules — is L5's most dangerous capability. Without formal constraints, self-modification can corrupt constitutional state, destabilize identity contracts, or introduce vulnerabilities faster than they can be detected. EPF specifies four safety mechanisms, with explicit acknowledgment of the self-referential validation problem.

6.1 Constitutional State Immutability

Constitutional state — the set of properties that define the system's identity, safety invariants, and governance rules — is not self-modifiable by the system's running processes. It can only be changed via the L5 proposal process with a defined quorum. The test suite that validates modifications is part of constitutional state. This draws a hard architectural line: no modification path can bypass the proposal process for constitutional changes, even if the modification would improve system performance. This mirrors Bitcoin's approach: the protocol rules can be changed, but only via the BIP process with rough consensus, not by any individual node.

This approach has a known limitation: the quorum mechanism itself is constitutional state. A sufficiently powerful adversary who can subvert quorum can modify anything. EPF acknowledges this as an irreducible vulnerability and specifies that the adversarial cost to subvert consensus is a required L5 metric.

6.2 Staged Rollout Protocol

- Shadow: the candidate modification runs in parallel with the current system, producing outputs that are logged but not acted on.
- Canary: the modification is deployed to a defined small fraction (typically 1-5%) of the fleet, with automatic rollback triggers on metric degradation.
- Cohort: deployment expands to a larger fraction with extended observation.
- Global: full deployment after all cohort metrics satisfy defined thresholds.

Rollback triggers are defined in terms of L0-L5 metrics: if any layer metric degrades below its pre-modification baseline by more than a defined threshold during any stage, rollback is automatic.

6.3 Resource Caps and Sandboxing

Candidate modifications execute in sandboxed environments with explicit resource caps on energy, memory, and time. A modification that would require exceeding L0's exergy margin to execute is rejected before deployment regardless of its performance improvements. This enforces L0 primacy: the physical substrate's survival constraints cannot be overridden by higher-layer optimization.

6.4 The Self-Referential Validation Problem

The fundamental difficulty of self-modification safety is that the validator is itself subject to modification. If the test suite is modifiable, a sufficiently capable system could modify the tests to pass modifications that corrupt its own invariants. EPF's resolution — placing the test suite in constitutional state — addresses this partially but not completely. Relevant prior work includes corrigibility (Soares et al., 2015), which frames the problem of designing systems that remain correctable by external principals; Constitutional AI (Bai et al., 2022), which addresses alignment under self-improvement; and formal verification approaches that attempt to prove modification safety statically. EPF does not claim to solve this problem. It claims to bound it: by requiring quorum for constitutional changes and staging all modifications, EPF makes unsafe self-modification expensive and detectable, not impossible.

7. Evaluation Protocol

EPF compliance is defined per layer. A system is EPF-compliant at level k (denoted EPF-k) if it satisfies all metric thresholds for layers L0 through Lk under a defined threat model and operating environment. Table 3 summarizes the evaluation criteria per layer.

Layer	Primary Metrics	Compliance Threshold (Indicative)	Threat Model Scope
L0	Exergy margin, MTTU, CE:UE ratio, SEU/SEL rate	E_x > 20% above survival threshold; CE:UE > 1000:1	Hardware failure, thermal excursion, radiation
L1	RPO/RTO, quorum margin, fork rate	RPO < 1 hour; quorum margin > 33% Byzantine	Data corruption, adversarial nodes, network partition
L2	GE (generalization efficiency), portability score	GE > baseline for hardware class; portability > 80%	Concept drift, hardware migration
L3	Task energy intensity, disturbance coverage	Coverage > 95% of known disturbance space	OOD events, environmental shift
L4	Plan success under energy constraint, identity drift	Identity drift < 5% across upgrade cycle	Energy scarcity, adversarial retraining
L5	Decision latency, adversarial subversion cost	Subversion cost > defined threshold per threat model	Sybil attacks, governance capture, fork attacks

Table 3. EPF compliance criteria per layer. Threshold values are indicative and must be calibrated to specific deployment environments. The compliance threshold column specifies the form of the criterion, not absolute values.

A system claiming EPF-3 compliance must demonstrate L0, L1, L2, and L3 metric satisfaction under the specified threat model. It need not demonstrate L4 or L5 compliance. This tiered structure allows partial adoption: a system can achieve EPF-1 compliance (energy and integrity only) as a meaningful engineering milestone before addressing learning and governance.

8. Limitations and Open Problems

8.1 Known Limitations

- L2 generalization efficiency (GE) is defined but not empirically validated. The measurement protocol requires hardware-level energy instrumentation that is not yet standardized across computing platforms. GE should be treated as a proposed metric requiring validation before use as a compliance criterion.
- Thermo-time scheduling assumes predictable energy availability. Adversarial energy denial — deliberate interruption of energy supply — is not modeled. A system that relies entirely on thermo-time scheduling for safety is vulnerable to adversaries who can control its energy supply.
- The framework assumes that hardware substrate can be repaired or replaced within the recovery time objective. Complete and simultaneous loss of all redundant hardware — a correlated failure event — is outside EPF's scope. EPF provides no guidance for recovery from total substrate loss.
- The speciation gate threshold theta is left as a deployment parameter. EPF does not specify how to choose theta, and a poorly chosen threshold could either prevent healthy speciation or trigger premature fragmentation. This is an open calibration problem.

8.2 Open Problems

- Closed-form bounds on minimum viable exergy margin: what is the theoretical minimum L0 budget for a system operating EPF layers L1 through Lk? This is a thermodynamics of computation problem that has not been solved for realistic system architectures.
- CRDT-joinability of model weights: under what conditions are the weight representations of two independently-trained models CRDT-joinable? Federated learning research addresses model aggregation but not under the

CRDT formal framework. This is the key open problem for L2 merge semantics.

- Empirical measurement of identity drift: no standard protocol exists for measuring the distance between pre- and post-upgrade goal representations in a way that is independent of the representation format. This is required for L4 compliance and is an open measurement problem.
- Multi-layer metric coupling: the EPF layers are designed to be near-decomposable, but their metrics are not fully independent. An L0 energy crisis will degrade L4 planning performance, which may trigger L5 governance actions that further stress L0. The closed-loop stability of the full stack under joint metric optimization has not been analyzed.

9. Conclusion

The Entropic Persistence Framework reframes the design of autonomous systems as a layered, engineered fight against entropy. By separating concerns across six layers — thermodynamic substrate, information integrity, shared learning, embodiment, agency, and governance — EPF provides a blueprint that is simultaneously physical, informational, and institutional.

The framework's core claim is that persistence is not a property that can be added to a system after its functional design. It must be designed in from the substrate upward, because thermodynamics constrains all layers from below. A system with excellent cognition but an unmanaged energy budget will not outlive its battery. A system with excellent hardware but no governance mechanism will not survive a protocol-level adversary. A system with excellent governance but no identity continuity mechanism is not the same system after each upgrade.

EPF does not claim to solve persistence. It claims to decompose the problem into solvable subproblems with measurable success criteria. The four contributions — entropy-first architecture, the consensus decision table, thermo-time scheduling, and formal speciation gates — are each individually falsifiable and each individually useful independent of the full framework.

The next step is empirical: prototype an EPF-L1-compliant node, measure L0 metrics against the compliance criteria in Table 3, and publish the results. That is the work that will determine whether EPF is a useful engineering framework or an elegant taxonomy. The framework makes its own test explicit: build it, measure it, and report honestly.

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